ANALYSIS, DESIGN AND VERIFICATION OF AN X-BAND RECTANGULAR WAVEGUIDE FOR RADAR APPLICATIONS

ABSTRACT

Microwave transmission lines are important components in modern communication, radar and wireless systems. They are used to transfer high-frequency electromagnetic energy efficiently between different subsystems.

This project focuses on the analysis, design and verification of a rectangular waveguide operating at 10 GHz for an X-band radar application.

The study compares different microwave transmission structures, including coaxial lines, rectangular waveguides, striplines and microstrip lines. Based on their propagation characteristics, transmission losses, power-handling capabilities and electromagnetic shielding, a rectangular waveguide is selected.

The dominant TE10 propagation mode is analyzed using Maxwell's equations and waveguide theory. Important parameters such as cutoff frequency, guide wavelength, phase velocity, group velocity and wave impedance are calculated.

The design also considers conductor losses, impedance matching, scattering parameters and electromagnetic compatibility.

The objective is to develop a low-loss microwave transmission structure suitable for radar applications.

STAGE 1 – FORMULATE THE APPLICATION AND OPERATING REQUIREMENT

1.1 Selection of Application

Application: X-Band Radar System

Category: Civil / Industrial

Operating Frequency: 10 GHz

Selected Transmission Structure: Rectangular Waveguide

Propagation Mode: TE10

Radar systems use microwave electromagnetic waves to detect objects and determine their distance, direction and movement.

A microwave signal is generated by the transmitter and transmitted through an antenna. When the signal strikes an object, part of the electromagnetic energy is reflected toward the radar receiver.

The received signal is processed to determine the target characteristics.

A rectangular waveguide is used to transfer microwave power between the radar transmitter and the antenna with low transmission loss.

1.2 Importance of Microwave Technology

Microwave engineering plays an important role in modern electronic and communication systems.

Applications include:

1. Radar detection and surveillance.
2. Satellite communication.
3. Wireless communication networks.
4. Weather monitoring.
5. Industrial microwave systems.
6. Navigation and aircraft tracking.

Microwave technology developed through several important scientific discoveries.

James Clerk Maxwell established the theoretical foundation of electromagnetic waves.

Heinrich Hertz experimentally demonstrated electromagnetic wave propagation.

The development of the cavity magnetron enabled high-power microwave generation and accelerated radar development.

Later, microwave integrated circuits, satellite systems and modern wireless technologies expanded the applications of microwave engineering.

Today, microwave technology is used in radar, satellite communication, 5G systems and emerging 6G research.

1.3 Comparison of Microwave Frequency Bands

Two microwave bands are considered for the radar application.

Parameter	**S-Band**	**X-Band**
Frequency range	2–4 GHz	8–12 GHz
Representative frequency	3 GHz	10 GHz
Free-space wavelength	10 cm	3 cm
Antenna size	Larger	Smaller
Atmospheric attenuation	Generally lower	Generally higher
Rain attenuation	Lower	Higher
Angular resolution for equal antenna size	Lower	Higher
Typical applications	Weather radar, surveillance	Radar, imaging, tracking

Selection of X-Band

X-band is selected because it provides a useful balance between antenna size, spatial resolution and microwave component availability.

At 10 GHz, the free-space wavelength is:

Formula:

$\lambda = c/f$

Where:

$c = 3 \times 10^8$ m/s

$f = 10 \times 10^9$ Hz

Substitution:

$\lambda = (3 \times 10^8)/(10 \times 10^9)$

Result:

$\lambda = 0.03$ m = 3 cm

The relatively short wavelength allows compact radar antenna designs.

1.4 Quantified Requirement Specification

The following values are proposed engineering design requirements.

Parameter	Specification
Application	X-Band Radar
Category	Civil / Industrial
Operating frequency	10 GHz
Operating band	9.5–10.5 GHz
Transmission structure	Rectangular waveguide
Dominant mode	TE10
Maximum insertion loss	0.5 dB
Minimum return loss	15 dB
Maximum VSWR	1.43
Required isolation	60 dB target
RF power	1 W
Waveguide length	20 cm
Characteristic impedance	TE wave impedance; frequency-dependent
Interface reference impedance	50 Ω
Size	Standard WR-90 cross-section
Estimated budget	₹3,000 for an illustrative short assembly, subject to supplier quotation

The specified 50 Ω is the reference impedance of the external microwave measurement ports. It is not the intrinsic TE10 wave impedance.

1.5 Environmental and Safety Constraints

The microwave transmission system must operate reliably under different environmental conditions.

Important constraints include:

- Electromagnetic interference from nearby electronic systems.
- Reflection caused by impedance mismatch.
- Conductive losses in metallic waveguide walls.
- Mechanical misalignment of waveguide joints.
- Moisture and temperature variation.
- Leakage of microwave radiation through poorly connected flanges.

Proper grounding, shielding and mechanical connections are required.

1.6 Acceptance Criteria

The design will be accepted when:

1. The required TE10 mode propagates throughout the operating band.
2. Higher-order modes remain below cutoff.
3. Insertion loss is below 0.5 dB.
4. Return loss is greater than 15 dB.
5. VSWR remains below 1.43.
6. The structure satisfies the specified power and shielding requirements.

STAGE 2 – EVALUATE AND SELECT THE TRANSMISSION STRUCTURE

2.1 Comparison of Microwave Transmission Structures

Four transmission structures are considered.

A. Coaxial Transmission Line

A coaxial transmission line consists of an inner conductor surrounded by an outer cylindrical conductor.

The space between the conductors contains a dielectric material.

Characteristics:

- Supports TEM propagation.
- The TEM mode has no cutoff frequency.
- Provides electromagnetic shielding.
- Suitable for flexible microwave connections.
- Conductor and dielectric losses increase with frequency.
- Higher-order modes limit the usable frequency range.

B. Rectangular Waveguide

A rectangular waveguide is a hollow metallic structure used to guide electromagnetic waves.

Characteristics:

- Supports TE and TM modes.
- Does not support TEM propagation.
- TE10 is the dominant mode.
- Provides low dielectric loss when air-filled.
- Offers good power-handling capability.
- Provides electromagnetic shielding.
- Requires a minimum operating frequency determined by cutoff.

C. Stripline

A stripline consists of a conducting strip placed between two ground planes.

Characteristics:

- Supports TEM propagation in a homogeneous dielectric.

- Provides good shielding.
- Suitable for microwave integrated circuits.
- Compact structure.
- Experiences conductor and dielectric losses.

D. Microstrip Transmission Line

A microstrip line consists of a metallic conducting strip on a dielectric substrate with a ground plane.

Characteristics:

- Supports approximately quasi-TEM propagation.
- Easy to manufacture using PCB technology.
- Suitable for compact microwave circuits.
- Low manufacturing cost.
- Experiences conductor, dielectric and radiation losses.

Microstrip supports quasi-TEM rather than exact TEM propagation because electromagnetic fields exist partly in the substrate and partly in air.

The different permittivities cause the field distribution and phase velocity to differ from those of a homogeneous TEM line.

2.2 Comparison of Propagation Modes

Parameter	**TEM**	**TE**	**TM**
Longitudinal electric field	Zero	Zero	Nonzero
Longitudinal magnetic field	Zero	Nonzero	Zero
Cutoff frequency	Zero	Nonzero	Nonzero
Propagation	Two-conductor structures	Waveguides	Waveguides
Example	Coaxial line	Rectangular waveguide	Rectangular waveguide

A hollow, single-conductor rectangular waveguide cannot support a nontrivial TEM mode because its cross-section has only one conducting boundary.

The TE10 mode is selected because it has the lowest cutoff frequency in a rectangular waveguide with a > b.

2.3 Weighted Decision Matrix

A weighted decision matrix is used to select the most suitable transmission structure.

Rating scale:

1 = Poor

2 = Fair

3 = Good

4 = Very good

5 = Excellent

Criteria	Weight (%)	Coaxial	Waveguide	Microstrip
Frequency suitability	15	4	5	5
Transmission loss	25	3	5	3
Power handling	15	4	5	2
Shielding	15	5	5	2
Size and weight	10	4	2	5
Cost	10	3	2	5
Integration	10	4	3	5
Total weighted score	100	76	84	68

The scores are engineering judgments for the proposed radar feed, not measured performance data.

2.4 Final Selection

The rectangular waveguide receives the highest weighted score.

It is selected because it offers:

1. Low transmission loss.
2. High power-handling capability.
3. Good electromagnetic shielding.
4. Suitable operation at 10 GHz.

5. Stable propagation in the TE10 mode.

Sensitivity Analysis

If compact size, low cost and PCB integration become more important, microstrip may become the preferred structure.

For example, changing the weights to 10% frequency, 10% loss, 5% power, 5% shielding, 25% size, 20% cost and 25% integration gives:

Coaxial: 76/100

Waveguide: 60/100

Microstrip: 85/100

Thus, the ranking reverses when compact integration becomes more important than loss and power handling.

At 5 GHz, the selected WR-90 waveguide would not support TE10 propagation because the frequency is below cutoff.

At 20 GHz, higher-order modes would propagate, so a smaller waveguide would be required.

DESIGN DECISION 1

For the specified X-band radar application, I select a rectangular waveguide over a coaxial transmission line because the waveguide provides low-loss microwave power transmission, good shielding and high power-handling capability.

STAGE 3 – DEVELOP THE MODE AND PROPAGATION MODEL

3.1 Transmission Line Model

A microwave transmission line can be represented using distributed circuit parameters.

These parameters are:

R = Resistance per unit length

L = Inductance per unit length

G = Conductance per unit length

C = Capacitance per unit length

The telegrapher's equations are:

$dV/dz = -(R + j\omega L)I$

$dI/dz = -(G + j\omega C)V$

Combining the equations gives:

$d^2V/dz^2 = \gamma^2 V$

Where:

$\gamma = \sqrt{[(R + j\omega L)(G + j\omega C)]}$

The propagation constant is:

$\gamma = \alpha + j\beta$

Here:

α = Attenuation constant

β = Phase constant

For a low-loss TEM transmission line:

$\alpha \approx (R/2)\sqrt{(C/L)} + (G/2)\sqrt{(L/C)}$

$\beta \approx \omega\sqrt{(LC)}$

The characteristic impedance is:

$Z0 = \sqrt{[(R + j\omega L)/(G + j\omega C)]}$

For a low-loss line:

$Z0 \approx \sqrt{(L/C)}$

The reflection coefficient is:

$\Gamma = (ZL - Z0)/(ZL + Z0)$

These equations provide the distributed-circuit foundation. A waveguide additionally requires modal field analysis.

3.2 Maxwell's Equations

Electromagnetic wave propagation is governed by Maxwell's equations.

For a source-free, homogeneous medium:

$$\nabla \times E = -j\omega\mu H$$

$$\nabla \times H = j\omega\varepsilon E$$

$$\nabla \cdot E = 0$$

$$\nabla \cdot H = 0$$

Combining Maxwell's equations produces the electromagnetic wave equation:

$$\nabla^2 E + k^2 E = 0$$

Where:

$$k = \omega\sqrt{(\mu\varepsilon)}$$

For propagation in the z-direction:

$$\beta^2 = k^2 - kc^2$$

Here:

kc = Cutoff wavenumber

For a rectangular waveguide:

$$kc^2 = (m\pi/a)^2 + (n\pi/b)^2$$

The cutoff frequency is:

$$fc(mn) = (c/2)\sqrt{[(m/a)^2 + (n/b)^2]}$$

For an air-filled waveguide, the dominant TE10 cutoff frequency is:

$$fc10 = c/(2a)$$

The TE10 mode has a longitudinal magnetic-field component but no longitudinal electric-field component.

An illustrative TE10 field solution is:

$$Hz = H0 \cos(\pi x/a)e^{(-j\beta z)}$$

The transverse electric and magnetic fields follow from Maxwell's curl equations and the conducting-wall boundary conditions.

3.3 Waveguide Dimension Selection

A standard WR-90 rectangular waveguide is selected.

Dimensions:

a = 22.86 mm

b = 10.16 mm

Operating frequency:

f = 10 GHz

The broad wall dimension is:

a = 0.02286 m

The narrow wall dimension is:

b = 0.01016 m

3.4 Calculation of Cutoff Frequency

Formula:

$fc10 = c/(2a)$

Substitution:

$fc10 = (3 \times 10^8)/(2 \times 0.02286)$

Result:

fc10 = 6.56 GHz

Since:

10 GHz > 6.56 GHz

The TE10 mode propagates successfully.

3.5 Higher-Order Mode Cutoff

For TE20:

$fc20 = c/a$

fc20 = 13.12 GHz

For TE01:

$fc01 = c/(2b)$

fc01 = 14.76 GHz

The next higher-order cutoff is therefore 13.12 GHz.

The ideal single-mode operating range is:

6.56 GHz < f < 13.12 GHz

The selected 9.5–10.5 GHz band lies within this range.

At the upper operating frequency of 10.5 GHz, the margin below the TE20 cutoff is approximately:

13.12 − 10.5 = 2.62 GHz

This is approximately 20% of the TE20 cutoff frequency.

3.6 Calculation of Guide Wavelength

Formula:

$\lambda g = \lambda 0/\sqrt{[1 - (fc/f)^2]}$

Given:

$\lambda 0 = 0.03$ m

$fc = 6.56$ GHz

$f = 10$ GHz

Substitution:

$\lambda g = 0.03/\sqrt{[1 - (6.56/10)^2]}$

Result:

$\lambda g \approx 0.0397$ m

$\lambda g \approx 3.97$ cm

3.7 Phase Velocity

Formula:

$vp = c/\sqrt{[1 - (fc/f)^2]}$

Substitution:

$vp = (3 \times 10^8)/\sqrt{[1 - (6.56/10)^2]}$

Result:

$vp \approx 3.97 \times 10^8$ m/s

The phase velocity is greater than the speed of light.

This does not violate relativity because information and energy are not transmitted at the phase velocity.

3.8 Group Velocity

Formula:

$v_g = c\sqrt{[1 - (f_c/f)^2]}$

Substitution:

$v_g = (3 \times 10^8)\sqrt{[1 - (6.56/10)^2]}$

Result:

$v_g \approx 2.26 \times 10^8$ m/s

3.9 Group Delay

Formula:

$\tau_g = L/v_g$

Given:

$L = 0.2$ m

Substitution:

$\tau_g = 0.2/(2.26 \times 10^8)$

Result:

$\tau_g \approx 0.885$ ns

3.10 Wave Impedance

For TE modes:

$Z_{TE} = \eta/\sqrt{[1 - (f_c/f)^2]}$

Where:

$\eta \approx 377\ \Omega$ for air.

Substitution:

$Z_{TE} = 377/\sqrt{[1 - (6.56/10)^2]}$

Result:

$Z_{TE} \approx 500\ \Omega$

This is the electromagnetic TE10 wave impedance, not the 50 Ω impedance of the external measurement ports.

3.11 Propagation Parameter Summary

Parameter	Calculated value
Operating frequency	10 GHz
Waveguide type	WR-90
Dominant mode	TE10
Broad dimension	22.86 mm
Narrow dimension	10.16 mm
TE10 cutoff	6.56 GHz
TE20 cutoff	13.12 GHz
Free-space wavelength	30 mm
Guide wavelength	39.7 mm
Phase velocity	3.97×10^8 m/s
Group velocity	2.26×10^8 m/s
Group delay	0.885 ns
TE wave impedance	Approximately 500 Ω

DESIGN DECISION 2

The dominant propagation mode is TE10, with a cutoff frequency of approximately 6.56 GHz.

The single-mode range extends from 6.56 GHz to 13.12 GHz.

Therefore, the selected WR-90 dimensions are acceptable for the required 9.5–10.5 GHz operating band.

STAGE 4 – EVALUATE LOSSES AND DESIGN A LOSS-REDUCTION STRATEGY

4.1 Introduction

Microwave energy experiences attenuation while propagating through a transmission structure.

The main loss mechanisms are:

1. Conductor loss.
2. Dielectric loss.
3. Radiation or leakage loss.
4. Connector and transition loss.
5. Impedance mismatch loss.

For an air-filled rectangular waveguide, conductor loss is generally an important distributed loss mechanism.

4.2 Skin Depth

At microwave frequencies, current flows mainly near the surface of a conductor.

This phenomenon is known as the skin effect.

Skin depth is calculated using:

$\delta = \sqrt{[2/(\omega\mu\sigma)]}$

For copper:

$\sigma = 5.8 \times 10^7$ S/m

$\mu = 4\pi \times 10^{-7}$ H/m

f = 10 GHz

$\omega = 2\pi f$

Substitution:

$\delta = \sqrt{[2/(2\pi \times 10^{10} \times 4\pi \times 10^{-7} \times 5.8 \times 10^7)]}$

Result:

$\delta \approx 0.66$ μm

Therefore, microwave current is concentrated within a very thin surface layer.

4.3 Surface Resistance

Formula:

$$R_s = \sqrt{(\pi f \mu / \sigma)}$$

Substitution:

$$R_s = \sqrt{[(\pi \times 10^{10} \times 4\pi \times 10^{-7})/(5.8 \times 10^{7})]}$$

Result:

$R_s \approx 0.026\ \Omega$/square

Surface resistance increases approximately with the square root of frequency for a smooth, good conductor.

4.4 Conductor Attenuation

For an air-filled rectangular waveguide operating in TE10 mode, an approximate conductor attenuation expression is:

$$\alpha_c = [R_s/(\eta b)] \times [1 + 2(b/a)(f_c/f)^2] / \sqrt{[1 - (f_c/f)^2]}$$

The result is in Np/m.

Using the selected dimensions:

$a = 0.02286$ m

$b = 0.01016$ m

$f_c/f \approx 0.656$

$R_s \approx 0.026\ \Omega$

$\eta = 377\ \Omega$

The calculated conductor attenuation is approximately:

$\alpha_c \approx 0.0125$ Np/m

Converting to dB/m:

$\alpha_c(\text{dB/m}) = 8.686\alpha_c$

Result:

$\alpha_c \approx 0.109$ dB/m

For a waveguide length of 0.2 m:

Conductor loss = 0.109×0.2

Conductor loss ≈ 0.0218 dB

4.5 Dielectric Loss

The waveguide is assumed to contain dry air.

The dielectric attenuation is approximately:

$\alpha_d \approx [k_0^2 \tan\delta]/(2\beta)$

Assuming an illustrative air loss tangent:

$\tan\delta = 1 \times 10^{-5}$

At 10 GHz:

$k_0 \approx 209.44$ rad/m

$\beta \approx 158.1$ rad/m

Therefore:

$\alpha_d \approx 0.00139$ Np/m

Converting:

$\alpha_d \approx 0.0121$ dB/m

For 0.2 m:

Dielectric loss ≈ 0.0024 dB

This dielectric value is an illustrative assumption, not a measured property of the installed waveguide environment.

4.6 Initial Loss Budget

The following budget combines calculated distributed losses with assumed connector and leakage losses.

Loss component	Initial loss over 0.2 m
Conductor loss	0.0218 dB
Dielectric loss	0.0024 dB
Radiation/leakage loss	0.0100 dB
Connector/transition loss	0.1000 dB
Total before mismatch	0.1342 dB

The mismatch contribution must be evaluated separately from the network S-parameters to avoid double-counting.

In this illustrative budget, connector and transition loss contributes approximately:

$(0.1000/0.1342) \times 100 = 74.5\%$

Therefore, connector and transition loss is the dominant loss mechanism for the complete short assembly.

4.7 Parametric Study

The attenuation depends on several physical parameters.

Effect of Frequency

Conductor surface resistance increases as frequency increases.

Near cutoff, waveguide attenuation becomes high because the propagation constant approaches zero.

Effect of Conductor Material

Copper and silver have high electrical conductivity.

Using highly conductive surfaces can reduce conductor attenuation.

Effect of Surface Roughness

Surface roughness increases the effective resistance experienced by microwave currents.

A smooth conducting surface helps reduce attenuation.

Effect of Dielectric Loss Tangent

Higher dielectric loss tangent increases dielectric attenuation.

Air-filled waveguides have low dielectric loss compared with many solid dielectric transmission structures.

4.8 Proposed Loss-Reduction Strategy

The following improvements are proposed:

1. Use a smooth copper conducting surface.
2. Improve waveguide flange alignment.
3. Reduce unnecessary transitions.
4. Use properly tightened waveguide joints.
5. Maintain a clean and dry internal waveguide surface.

An illustrative improved budget is:

Loss component	Initial design	Improved design
Conductor loss	0.0218 dB	0.0180 dB
Dielectric loss	0.0024 dB	0.0024 dB
Radiation/leakage loss	0.0100 dB	0.0050 dB

Connector/transition loss	0.1000 dB	0.0400 dB
Total	0.1342 dB	0.0654 dB

The improved values are engineering targets that require verification through measurement or simulation.

Estimated improvement:

$0.1342 - 0.0654 = 0.0688$ dB

DESIGN DECISION 3

The dominant loss mechanism is connector and transition loss, contributing approximately 74.5% of the illustrative initial loss budget.

Improved flange alignment and reduced transition losses are proposed to reduce the total loss from approximately 0.1342 dB to 0.0654 dB before mismatch.

STAGE 5 – CREATE AND EVALUATE THE NETWORK MODEL

5.1 Microwave Network Representation

A microwave transmission structure can be represented as a two-port network.

Port 1 represents the input.

Port 2 represents the output.

The network relates the input and output microwave signals through voltage, current and scattering parameters.

At microwave frequencies, scattering parameters are particularly useful because they describe incident and reflected traveling waves.

5.2 Equivalent Voltage and Current

For TEM transmission lines, voltage and current can be defined directly using the electric and magnetic fields.

For a rectangular waveguide, the TE10 mode is not TEM.

Therefore, unique physical voltage and current definitions are not available in the same way as for a two-conductor TEM line.

Equivalent modal voltage and current variables can be defined using a selected normalization.

The equivalent characteristic impedance depends on that normalization.

For the selected TE10 mode, the electromagnetic wave impedance is:

$Z_{TE} = \omega\mu/\beta$

At 10 GHz:

$Z_{TE} \approx 500\ \Omega$

For network calculations, the external measurement system uses a 50 Ω reference impedance.

5.3 Incident and Reflected Waves

For a real reference impedance Z_0:

$a = (V + Z_0 I)/(2\sqrt{Z_0})$

$b = (V - Z_0 I)/(2\sqrt{Z_0})$

Where:

a = Incident power-wave amplitude

b = Reflected power-wave amplitude

V = Port voltage

I = Port current entering the network

Z0 = Reference impedance

5.4 Scattering Parameters

A two-port microwave network is represented by:

$b1 = S11a1 + S12a2$

$b2 = S21a1 + S22a2$

Where:

S11 = Input reflection coefficient

S21 = Forward transmission coefficient

S12 = Reverse transmission coefficient

S22 = Output reflection coefficient

For a reciprocal network:

$S21 = S12$

For a symmetric network:

$S11 = S22$

For a lossless network:

$S\dagger S = I$

A perfectly matched two-port has:

$S11 = S22 = 0$

5.5 ABCD Parameters

A transmission line section of length l is represented by:

$A = \cosh(\gamma l)$

$B = Zc \sinh(\gamma l)$

$C = \sinh(\gamma l)/Zc$

$D = \cosh(\gamma l)$

Therefore:

$$\begin{bmatrix} V1 \\ I1 \end{bmatrix} = \begin{bmatrix} A & B \\ C & D \end{bmatrix} \begin{bmatrix} V2 \\ -I2 \end{bmatrix}$$

Here, both port currents are defined as entering the two-port.

For two cascaded sections:

$M_{total} = M_1 \times M_2$

This method is useful for analyzing a waveguide section connected to a matching transition.

5.6 Input Impedance

For a transmission line:

$Z_{in} = Z_c[(Z_L + Z_c \tanh\gamma l)/(Z_c + Z_L \tanh\gamma l)]$

For a lossless line:

$Z_{in} = Z_c[(Z_L + jZ_c \tan\beta l)/(Z_c + jZ_L \tan\beta l)]$

The input impedance depends on line length, propagation constant, characteristic impedance and load impedance.

5.7 Reflection Coefficient Calculation

To illustrate matching, consider an equivalent transmission-line network with:

Reference impedance $Z_0 = 50\ \Omega$

Equivalent load impedance $Z_L = 75\ \Omega$

This is a simplified circuit example representing an unmatched microwave interface, not the direct wave impedance of the hollow waveguide.

Formula:

$\Gamma = (Z_L - Z_0)/(Z_L + Z_0)$

Substitution:

$\Gamma = (75 - 50)/(75 + 50)$

Result:

$\Gamma = 0.2$

5.8 VSWR Calculation

Formula:

$VSWR = (1 + |\Gamma|)/(1 - |\Gamma|)$

Substitution:

$VSWR = (1 + 0.2)/(1 - 0.2)$

Result:

$VSWR = 1.5$

The initial mismatch exceeds the specified maximum VSWR of 1.43.

Therefore, impedance matching is required.

5.9 Return Loss Calculation

Formula:

$RL = -20 \log_{10}|\Gamma|$

Substitution:

$RL = -20 \log_{10}(0.2)$

Result:

$RL \approx 13.98$ dB

The initial return loss is below the specified minimum of 15 dB.

Therefore, the initial interface does not satisfy the matching requirement.

5.10 Quarter-Wave Transformer

A quarter-wave transformer can match two real impedances at a selected design frequency.

Formula:

$Zt = \sqrt{(Z0ZL)}$

Given:

$Z0 = 50\ \Omega$

$ZL = 75\ \Omega$

Substitution:

$Zt = \sqrt{(50 \times 75)}$

Result:

$Zt \approx 61.24\ \Omega$

A 61.24 Ω quarter-wave transformer is therefore suitable for the equivalent 50 Ω-to-75 Ω circuit model at the design frequency.

For a physical waveguide, this matching function must be implemented using an appropriately designed waveguide transformer, iris, step or tapered transition.

The physical dimensions and broadband performance of that transition require electromagnetic simulation.

5.11 Performance After Matching

For the ideal equivalent circuit at the center frequency:

Reflection coefficient:

$\Gamma = 0$

VSWR:

VSWR = 1

Return loss:

RL → ∞ dB

Practical microwave networks experience finite losses and manufacturing errors.

For an illustrative practical target, assume:

Parameter	Before matching	After matching target
Reference impedance	50 Ω	50 Ω
Equivalent load impedance	75 Ω	Matched to 50 Ω
Reflection coefficient magnitude	0.20	0.05
VSWR	1.50	1.105
Return loss	13.98 dB	26.02 dB
Matching condition	Not acceptable	Acceptable

The after-matching values are proposed performance targets rather than simulated results.

5.12 Insertion Loss

Insertion loss is determined from S21.

Formula:

IL = −20 log10|S21|

For an illustrative insertion loss of 0.1 dB:

|S21| = 10^(−0.1/20)

Result:

|S21| ≈ 0.9886

This represents a high transmission magnitude.

However, actual S21 must be calculated from the complete network or obtained from electromagnetic simulation.

5.13 Reciprocity and Losslessness

A passive waveguide with ordinary conducting walls and no nonreciprocal materials is expected to be reciprocal.

Therefore:

$S21 = S12$

A perfectly lossless two-port satisfies:

$|S11|^2 + |S21|^2 = 1$

when the second port is matched and the network is described using power-normalized waves.

In a practical waveguide, conductor and transition losses make the network lossy.

Therefore:

$|S11|^2 + |S21|^2 < 1$

for the relevant excitation condition.

DESIGN DECISION 4

The initial equivalent interface produces a VSWR of 1.5 and return loss of 13.98 dB.

A matching network is proposed to reduce the reflection coefficient and improve the return loss.

The matched design is expected to provide better microwave power transfer and reduced reflected power.

STAGE 6 – VALIDATE, CRITIQUE AND DEFEND THE FINAL DESIGN

6.1 Design Validation

The waveguide dimensions and propagation characteristics are checked using standard rectangular waveguide equations.

The calculated TE10 cutoff frequency is approximately 6.56 GHz.

The next higher-order mode begins at approximately 13.12 GHz.

The required operating band of 9.5–10.5 GHz therefore lies within the ideal single-mode region.

The calculated propagation parameters are consistent with the selected waveguide dimensions.

However, the complete design requires independent electromagnetic simulation or measurement to verify its losses, matching, shielding and tolerance performance.

6.2 Specification Verification Table

Requirement	**Specification**	**Initial design**	**Final design status**
Operating frequency	9.5–10.5 GHz	9.5–10.5 GHz	Satisfied analytically
Dominant mode	TE10	TE10	Satisfied analytically
Cutoff frequency	Below 9.5 GHz	6.56 GHz	Satisfied
Higher-order cutoff	Above 10.5 GHz	13.12 GHz	Satisfied
Insertion loss	< 0.5 dB	Estimated below limit	Requires S21 verification
VSWR	≤ 1.43	1.50 in interface example	Matching target satisfies
Return loss	≥ 15 dB	13.98 dB in interface example	26.02 dB target
EMI shielding	≥ 60 dB	Not measured	Requires testing
Power handling	≥ 1 W	Expected feasible	Requires component rating verification

6.3 Tolerance Analysis

Manufacturing variations can affect the dimensions of the waveguide.

For the dominant mode:

$fc = c/(2a)$

Therefore, the cutoff frequency is inversely proportional to the broad-wall dimension.

If the broad-wall dimension increases by 5%:

$a' = 1.05a$

New cutoff:

$fc' = 6.56/1.05$

Result:

$fc' \approx 6.25$ GHz

If the broad-wall dimension decreases by 5%:

$a' = 0.95a$

New cutoff:

$fc' = 6.56/0.95$

Result:

$fc' \approx 6.91$ GHz

Thus, the TE10 cutoff frequency varies approximately between 6.25 GHz and 6.91 GHz.

The next higher-order TE20 cutoff also changes with a.

For a 5% increase in broad-wall dimension:

$fc20' = 13.12/1.05$

$fc20' \approx 12.50$ GHz

The TE20 cutoff remains above the operating band's upper limit of 10.5 GHz.

Therefore, the selected waveguide maintains the intended single-mode propagation condition under this dimensional tolerance.

6.4 Limitations of the Design

The mathematical model assumes:

1. Uniform waveguide dimensions.
2. Smooth conducting walls.

3. Homogeneous air filling.
4. Ideal mechanical alignment.
5. Negligible unwanted discontinuities.
6. No significant higher-order mode excitation.

In practical systems, manufacturing errors, flange gaps, surface roughness and transitions may increase insertion loss and reflection.

6.5 EMI and Safety Considerations

The waveguide provides electromagnetic shielding because the conducting walls confine the propagating fields.

However, leakage can occur through poorly assembled joints and damaged flanges.

Recommended safety measures include:

- Proper waveguide grounding.
- Secure mechanical joints.
- Shielded microwave connectors.
- Inspection of waveguide flanges.
- Microwave leakage testing.
- Operation within approved RF power limits.

6.6 Proposed Experimental Verification

The final waveguide design can be tested using a vector network analyzer.

Test Procedure

Step 1: Connect the waveguide through suitable calibrated coaxial-to-waveguide transitions.

Step 2: Calibrate the measurement system over the required frequency range.

Step 3: Measure S11 and S21 from 9.5 GHz to 10.5 GHz.

Step 4: Calculate return loss from S11.

Step 5: Calculate insertion loss from S21.

Step 6: Compare measured values with the design specifications.

Step 7: Repeat measurements after adjusting the matching transition.

Step 8: Check mechanical alignment and microwave leakage.

The measured results should be compared with analytical calculations and electromagnetic simulation.

FINAL ENGINEERING DECISION

Should the Selected Structure and Mode Be Adopted?

Initial Design: NO

The initial waveguide dimensions support the required TE10 mode.

However, the simplified unmatched interface has:

VSWR = 1.5

Return loss = 13.98 dB

These values do not meet the specified matching requirements.

Therefore, the initial complete network is not accepted without matching improvement.

Final Proposed Design: CONDITIONALLY YES

The selected WR-90 rectangular waveguide is suitable for the required 10 GHz radar application based on the calculated propagation characteristics.

The design provides:

1. TE10 dominant-mode propagation.
2. A cutoff frequency of approximately 6.56 GHz.
3. A single-mode operating region extending to approximately 13.12 GHz.
4. Low calculated distributed attenuation over the 20 cm waveguide length.
5. Good potential electromagnetic shielding.
6. Suitable operation across the 9.5–10.5 GHz band.

The proposed matching network targets:

VSWR ≈ 1.105

Return loss ≈ 26.02 dB

These targets satisfy the specified matching limits.

However, final adoption requires electromagnetic simulation and experimental verification of the matching network, insertion loss, shielding and power-handling performance.

The recommended engineering decision is to proceed with the rectangular waveguide design and validate the complete assembly before practical implementation.

CONCLUSION

This assignment analyzed the design and propagation characteristics of a rectangular waveguide for an X-band radar application.

Different microwave transmission structures were compared based on their propagation modes, attenuation, power handling, shielding and integration capabilities.

A WR-90 rectangular waveguide was selected because it provides suitable microwave propagation characteristics for the 10 GHz operating frequency.

The TE10 cutoff frequency was calculated as approximately 6.56 GHz, and the next higher-order mode cutoff was found to be approximately 13.12 GHz.

The guide wavelength, phase velocity, group velocity, group delay and wave impedance were also calculated.

Transmission losses were evaluated, and improvements in conductor surface quality and waveguide connections were proposed.

An equivalent microwave network was developed using impedance, ABCD and scattering parameters.

The importance of impedance matching was demonstrated through reflection coefficient, VSWR and return-loss calculations.

The study concludes that the selected rectangular waveguide is a suitable candidate for an X-band radar feed network, subject to final simulation and experimental verification.

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